Simulation of a Multi-Target Pulse-Doppler Radar Receiver with One- and Two-Dimensional CFAR Detection

Technical Report

Daksh Shah
Third-year B.Tech, Department of Electrical Engineering
National Institute of Technology Rourkela, India

Repository: https://github.com/PLANETORIAN/Pulse_Radar · MATLAB/Simulink R2026a · 15 October 2026

Seeds: thermal noise `current_seed`+0, 1; clutter +2, 3; target k +100k; CFAR tests `rng(1)`, then `randi(100000,N,1)`

TABLE I
KEY RESULTS: TRUE, MEASURED AND ERROR (SIMULINK, THERMAL NOISE ONLY; CA/OS: CELL-AVERAGING/ORDER-STATISTIC CFAR)

Quantity	True	Measured	Error (meas. − true)
Range (km), both 2-D CFARs	20/40/70	19.95/40.05/70.05	−0.05/+0.05/+0.05
Velocity (m/s), OS	2/3/4	2.017/2.999/4.021	+0.017/−0.001/+0.021
Velocity (m/s), CA	2/3/4	1.967/3.049/4.014	−0.033/+0.049/+0.014
All three found together	3 targets	OS 46/50, CA 10/50	–
Clusters matching no target	0	OS 5, CA 58 (110 trials, 4 cases)	–
Doppler loss at 200 m/s*	−9.21 dB (computed)	−9.17 dB	+0.04 dB
Alias of 200 m/s*	5.0 m/s	5.031 m/s (pulse-pair)	+0.031 m/s

Velocities from one frame. *Single target; loss re. 5 m/s, Hamming filter; true alias = 200 mod 15.

***Abstract*—This report documents a simulation-only pulse-Doppler radar receiver, built in MATLAB/Simulink, for several simultaneous point targets at 10 GHz. Echo amplitudes follow the radar range equation with independent Swerling I fluctuation, thermal noise follows a kT_0BF budget, and target delays are applied as frequency-domain phase ramps. Two receiver paths are compared. A one-dimensional chain (moving-target indication, a Hamming-weighted matched filter, cell-averaging constant false-alarm-rate detection, peak logic, range and pulse-pair velocity) handles a single return only. A 16-pulse range–Doppler map with two-dimensional cell-averaging and order-statistic detection and connected-component centroiding reports every target: 19.95, 40.05 and 70.05 km, with velocities within 0.05 m/s. Order-statistic detection found all three targets in 46 of 50 trials against 10 of 50 for cell-averaging detection. False-alarm rates have not been validated by a noise-only simulation, all results use thermal noise only, and the unmodulated pulse limits both range resolution and blind range to 15 km.**

I. INTRODUCTION

A pulsed radar transmits a short burst, listens for the echo, and infers range from the delay and radial velocity from the change of the echo's phase between pulses. This report describes a Simulink model of such a receiver at 10 GHz

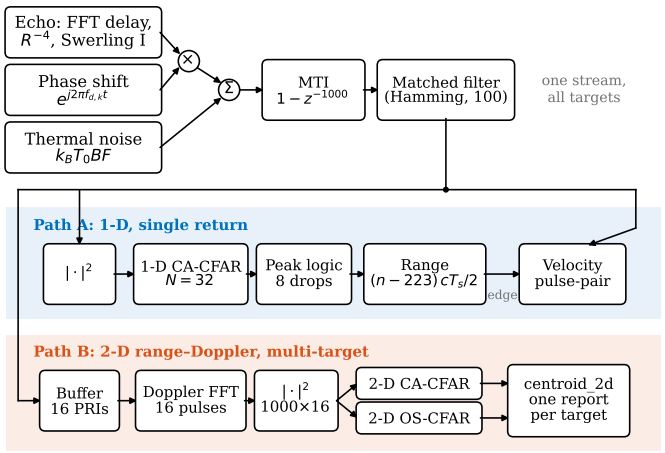


Fig. 1. Block diagram of `Pulse_Radar.slx` (redrawn). The per-target echoes and Doppler phasors are multiplied, summed with thermal noise and filtered into one stream. Path A is the one-dimensional chain; Path B builds the range–Doppler map, runs both two-dimensional CFARs and clusters the detections. The coloured-clutter generator, which is not connected in the signal path, is omitted here (see Sec. III).

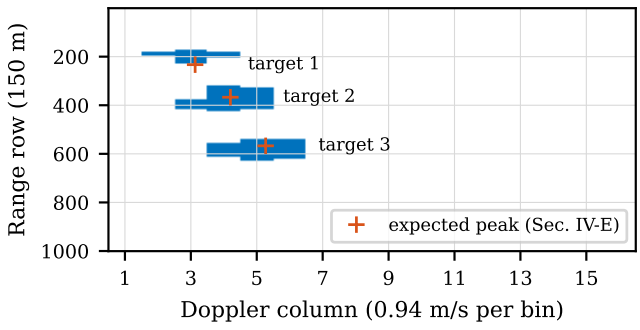


Fig. 2. Two-dimensional CA-CFAR detection map (frame 2) for targets at 20, 40 and 70 km moving at 2, 3 and 4 m/s; blue cells are detections, crosses the expected peaks. The 1000 × 16 map is read back from the Simulink Matrix Viewer capture (a different run from Sec. IV-F). Each target yields a block of detections, which Path B reduces to one report.

($\lambda = 3$ cm) with a 1 ms pulse repetition interval (PRI) and a 100 μs unmodulated pulse. Everything is simulated; no hardware or recorded data is used. The contributions are:

- a physically scaled scene: radar-range-equation amplitudes, independent Swerling I fluctuation [1] of each target's radar cross section (RCS) and a kT_0BF noise budget that sets the signal-to-noise ratio (SNR);
- a one-dimensional receiver (Path A: moving-target indication (MTI), matched filter, detection, range, velocity) and a multi-target range–Doppler receiver (Path B) with cell-averaging (CA) and order-statistic (OS) constant false-alarm-rate (CFAR) detection [2], [3] and clustering;
- analytical checks of every derived quantity, a noise-only analysis of the 1-D false-alarm rate, and paired Monte Carlo comparisons of the two CFARs.

Fig. 1 shows the top level of the model. Path B exists because Path A cannot separate targets: its range and velocity blocks hold the most recent trigger, so with three targets they report one value. The derivations behind each result are collected in Appendix A.

II. Radar Fundamentals and Derived System Limits

Assumptions: baseband in-phase/quadrature (I/Q) samples at $T_s = 1$ μs, complex white Gaussian noise, point targets at constant radial velocity, one coherent processing interval (CPI) of 16 pulses. Table VII (Appendix D) lists the limits that follow from the parameters in `init.m`; each was recomputed in `checks.py`.

Range. An echo must return before the next pulse is sent, otherwise it is credited to the wrong pulse:

$$R_u = \frac{c\,\mathrm{PRI}}{2} = 150 \text{ km}, \tag{1}$$

where $c = 3 \times 10^8$ m/s and PRI $= 1$ ms. A target at 160 km therefore appears at 10 km (range folding); the frequency-domain echo model reproduces this by a circular shift (Sec. III). While the transmitter is on the receiver is blanked, so echoes from $R < c\tau_p/2 = 15$ km are partly eclipsed; Fig. 7 (Appendix C) shows that a 10 km target then appears 0.9 dB low and displaced to 12.5 km. Blanking is not modelled in Simulink, so this result is analytical only. The 150 m bin spacing is only the sampling grid: two point targets merge unless they are separated by the pulse length, $c\tau_p/2 = 15$ km. Sampling 100 times faster than the 10 kHz pulse bandwidth does not change this; a linear-FM chirp of bandwidth B would give $c/(2B)$ at the same pulse energy [4].

Doppler and its two timescales. A target closing at v rotates the baseband phase at

$$f_d = \frac{2v}{\lambda}, \tag{2}$$

where $\lambda = 0.03$ m. Within one pulse the phasor turns $f_d\tau_p$ times; at 200 m/s that is 1.33 turns, so a filter that assumes constant phase partly cancels itself. Fig. 3 compares the measured sweep with the computed loss: -9.17 dB measured against -9.21 dB computed for the Hamming filter at 200 m/s, and -12.43 dB for a rectangular filter. Between pulses the phasor turns f_d PRI times, but only the fraction of a turn is observable, like a wagon wheel filmed at 1000 frames per second. Velocities are unambiguous within $\pm\lambda/(4\,\mathrm{PRI}) = \pm 7.5$ m/s and repeat every 15 m/s, so 200 m/s (13.33 turns) is read as 5.0 m/s (Fig. 9, Appendix C). Raising the pulse repetition frequency (PRF) widens the velocity window but shrinks R_u; this is the PRF dilemma that staggered PRFs resolve [5].

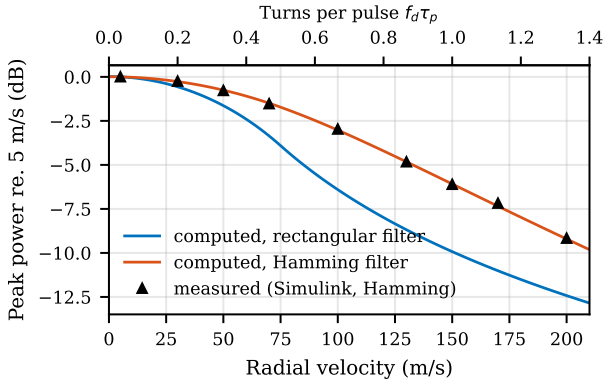


Fig. 3. Within-pulse Doppler mismatch loss. Lines: computed peak of the matched-filter output over all lags (analytical). Triangles: Simulink sweep with constant RCS and fixed seed. The Hamming filter loses 3.2 dB less than a rectangular one at 200 m/s.

TABLE II
Computed SNR for $\sigma = 1$ m^2 (Not Measured)

R (km)	P_r (dBm)	N_{delay} (samples)	SNR/sample (dB)	+ matched filter (dB)	+ 16-pulse FFT (dB)
20	-87.5	133	23.5	42.1	52.6
40	-99.5	267	11.5	30.1	40.6
70	-109.2	467	1.7	20.4	30.9

Signal-to-noise ratio. The received power of target k and the thermal noise power are

$$P_{r,k} = \frac{P_t G^2 \lambda^2 \sigma_k}{(4\pi)^3 R_k^4 L}, \qquad P_n = k_B T_0 B F, \tag{3}$$

where $P_t = 100$ kW, $G = 35$ dB, σ_k is the mean RCS (1 m^2), R_k the range, $L = 2$ dB the system loss, $k_B = 1.38 \times 10^{-23}$ J/K, $T_0 = 290$ K, $B = 1$ MHz and $F = 3$ dB, giving $P_n = 7.99 \times 10^{-15}$ W (-111.0 dBm). Table II gives the computed values; the Hamming Doppler window costs 1.54 dB of the 12.04 dB coherent gain of 16 pulses. Fig. 5 (Appendix C) shows the same as a function of range.

III. Simulation Model

The *Echo* subsystem buffers one PRI (1000 samples), applies the FFT, multiplies by one phase ramp per target, $H_k[n] = e^{-j2\pi n N_{d,k}/1000}$ with $N_{d,k} = \mathrm{round}(2R_k/(cT_s))$, applies the inverse FFT and scales each column by $A_k = \sqrt{-\sigma_k c_k \ln U_k}$, where c_k is the received power for 1 m^2 and U_k is uniform on $(10^{-4}, 1)$. The draw is held for the 16 ms run, so each target fluctuates independently from run to run (Swerling I). The *Phase Shift* block multiplies channel k by

$e^{j2\pi f_{d,k}t}$, and the channels are summed with complex thermal noise. A run is seeded by `current_seed`: the noise uses offsets 0–3 and target k uses $100k$. Because the buffer must fill first, every echo leaves the chain exactly one PRI late, so the first PRI holds no echo (Fig. 12, Appendix C); a whole-PRI delay leaves the position within the PRI, and hence every range calculation, unchanged. A target is switched off with $\sigma_k = 0$, and range is floored at 1 m so that $c_k \propto R^{-4}$ stays finite in single precision (Sec. V). The coloured-clutter generator (50× the thermal variance, 100-tap smoothing filter) was built and simulated on its own and is not connected, so all end-to-end results use thermal noise only.

IV. Receiver Chain

A. Matched Filter

The filter correlates the received signal with the expected pulse. For a 100-sample rectangle the matched taps are all ones; the model weights them with a Hamming window $w[k]$ [6]:

$$y[n] = \sum_{k=0}^{99} w[k]\, x[n-k], \qquad G_{\mathrm{MF}} = \frac{(\sum w)^2}{\sum w^2}, \tag{4}$$

where x is the input and G_{MF} the SNR gain: 18.6 dB against 20.0 dB for ones, a 1.4 dB cost. The window is symmetric, so the peak stays at 99 samples after the echo start and the output still spans 199 samples. The benefit is lower range sidelobes and the better Doppler tolerance of Fig. 3. Pitfall: any filter change alters the shape that the downstream CFAR guard and peak logic were tuned for (Sec. V). Fig. 6 (Appendix C) shows the square-law output for three targets.

B. Moving-Target Indication (MTI)

A two-pulse canceller subtracts the previous PRI, $H(z) = 1 - z^{-1000}$, so returns that do not change between pulses cancel:

$$|H(f_d)|^2 = 4\sin^2(\pi f_d\,\mathrm{PRI}), \tag{5}$$

where f_d is the target's Doppler shift and PRI $= 1$ ms. The response is zero at the blind speeds $k \times 15$ m/s (Fig. 8, Appendix C). For the targets at 2, 3 and 4 m/s the gains are -1.8, $+1.4$ and $+3.4$ dB. Validation: the uncancelled first peak in Fig. 6 is 1.9 dB above the steady state, against 1.8 dB predicted for target 1. A target moves only 2–4 mm per PRI, far less than the 150 m bin, so range migration does not degrade the subtraction. No clutter-residue measurement exists because the clutter path is disconnected; the MTI improvement factor is listed under future work (Sec. VII).

C. One-Dimensional CA-CFAR

For each cell under test (CUT) the threshold is α times the mean of $N = 32$ training cells, skipping 100 guard cells per side. If the noise power is exponential with unknown mean μ, the sum of the training cells is Gamma-distributed and

$$P_{fa} = \left(1 + \frac{\alpha}{N}\right)^{-N}, \tag{6}$$

where N is the number of training cells and α the threshold multiplier; μ cancels, which is what makes the rate constant (Appendix A). For $P_{fa} = 10^{-6}$, $\alpha = 17.28$ (12.37 dB), 0.97 dB above the known-noise value $\ln 10^6 = 13.82$. The averaging is a 233-tap FIR filter with the CUT delayed by 116 samples, and the guard covers the 199-sample response (scope capture in Fig. 13, Appendix C). While only $m < N$ training cells hold data, $P_{fa}(m) = (1 + \alpha/N)^{-m}$, which is 10^{-3} at $m = 16$. *Pitfall:* the derivation assumes independent cells, but matched-filter noise is correlated over 199 samples, so each 16-cell strip is close to one independent sample. A noise-only Python mirror of this path (no Simulink) gives $P_{fa} = 7.9\times10^{-3}$ with the Hamming filter and 5.0×10^{-3} with ones, against 8.2×10^{-7} for white noise (Fig. 10, Appendix C). This is an analytical prediction; the noise-only P_{fa} of the Simulink 1-D path has not been measured (Sec. VII).

D. Detection, Range and Velocity (Path A)

The peak logic fires once the CFAR mask is high and the signal has fallen for 8 consecutive samples, then ignores 150 samples ($1.5N_\tau$) and stretches the trigger to 10 samples for display (Fig. 14, Appendix C). The range block subtracts the pipeline latency $99 + 116 + 8 = 223$ samples:

$$R = (n_{\mathrm{trig}} - 223)\,\frac{cT_s}{2}, \tag{7}$$

where n_{trig} is the trigger index within the PRI; a trigger at 356 gives 133×150 m $= 19.95$ km. The pulse-pair estimator measures $\phi = \arg(x[n]x^*[n-1000]) = 4\pi v\,\mathrm{PRI}/\lambda$ and outputs $v = \lambda\phi/(4\pi\,\mathrm{PRI})$ (2.39 m/s per radian); 200 m/s gives $\phi = 2\pi/3$, i.e. 5.0 m/s, and 5.031 m/s was measured with a single target. Pitfalls: one sample pair, no averaging; a weak target's noisy hump may never fall for 8 samples in a row, and the SNR below which this happens has not been measured (Sec. VII); and both blocks hold the latest trigger, so Path A serves one target only.

E. Range–Doppler Map and 2-D CA-CFAR

The FFT subsystem stacks 16 PRIs into a 1000×16 matrix, applies a 16-point Hamming window across pulses (first sidelobe -39 dB against -13 dB) and takes

$$X[r,m] = \sum_{p=0}^{15} w[p]\, y_p[r]\, e^{-j2\pi mp/16}, \tag{8}$$

where $y_p[r]$ is matched-filter sample r of pulse p and bin m maps to $m \times 0.9375$ m/s. The 2-D CA-CFAR uses a ring of $T_r = 16$, $T_d = 1$ training and $G_r = 100$, $G_d = 3$ guard cells per side: $N = 233 \times 9 - 201 \times 7 = 690$ and $\alpha = 13.955$. The map is padded so that every row is tested: in range with $\mathrm{median}/\ln 2$ (the mean of exponential noise, insensitive to targets), in Doppler circularly, because bin 16 neighbours bin 1. Expected peaks are at rows 233, 367 and 567 and Doppler columns 3.13, 4.20 and 5.27, as observed in Fig. 2.

F. Clustering

`centroid_2d` joins touching detections (8-connected, wrapping in Doppler), takes range from the peak row, $(r_{\mathrm{pk}} - 100) \times 150$ m, and velocity from the power-weighted centroid column, which interpolates between the 0.94 m/s bins. It reports up to eight clusters and, as a headline, the strongest. Range comes from the peak because a mask cut on one side pulls the centroid: one CA cluster was 130 cells with its centroid 14 rows from its peak, while OS masks had 323–373 cells and centroids within 1.4 rows.

G. OS-CFAR

OS-CFAR sorts the $N = 626$ training cells ($G_d = 2$) and uses the $k = 470$th (75th percentile) as the noise estimate, so a neighbour covering under a quarter of the ring is ignored [3]. For exponential noise

$$P_{fa} = \prod_{j=N-k+1}^{N} \frac{j}{j+\alpha}, \tag{9}$$

where N is the number of training cells, k the rank used as the noise estimate, α the multiplier and j the product index. The model's $\alpha = k(P_{fa}^{-1/k} - 1) = 14.02$ is the CA formula evaluated with k; the exact value is 10.13, so the threshold is 1.41 dB conservative ($P_{fa} \approx 5.6 \times 10^{-9}$ ideally).

V. Design Issues and Resolutions

Fourteen issues were recorded during development, among them the matched-filter change that broke Path A, CA-CFAR masking and the 2-D CFAR edge fixes; Table IV in Appendix B gives each with its cause, fix and evidence.

VI. System-Level Performance

Baseline. With targets at 20, 40 and 70 km and 2, 3 and 4 m/s, both CFARs report 19.95, 40.05 and 70.05 km, the 150 m-quantised values of 133.3, 266.7 and 466.7 samples, with velocity errors under 0.05 m/s (Table I). In the same frame CA-CFAR adds a 17-cell cluster at 111 km that matches no target.

CA versus OS. `os_cfar_test.m` runs paired trials: both CFARs see the same noise and Swerling draws in each run. Clusters are matched one-to-one within 10 rows, and velocity error is taken modulo 15 m/s. Table III gives Wilson 95 % intervals [7]; the 20-trial cases are wide (a perfect 20/20 only bounds the rate above 83.9 %). Fig. 4 shows the per-target rates of all four cases. OS-CFAR recovers the masked middle target and produces 5 unmatched clusters against 58 over 110 trials. The two detectors also differ in geometry and threshold: CA-CFAR uses $G_d = 3$ (690 training cells) and OS-CFAR $G_d = 2$ (626 cells) with the conservative α of Sec. IV-G, so the unmatched-cluster comparison mixes the effect of the detector with that of the threshold. Neither separates two targets in one range cell two Doppler bins apart (case 4).

TABLE III
CA- Versus OS-CFAR, All Active Targets Found Together

Case	Scene, trials	CA	OS	Unmatched CA / OS
1	3 targets, 50	10/50 [11, 33]	46/50 [81, 97]	31 / 2
2	2 separated, 20	20/20 [84, 100]	20/20 [84, 100]	7 / 1
3	2 strong, 20	14/20 [48, 86]	20/20 [84, 100]	10 / 1
4	same cell, 20	strong target 20/20, weak target not separated		10 / 1

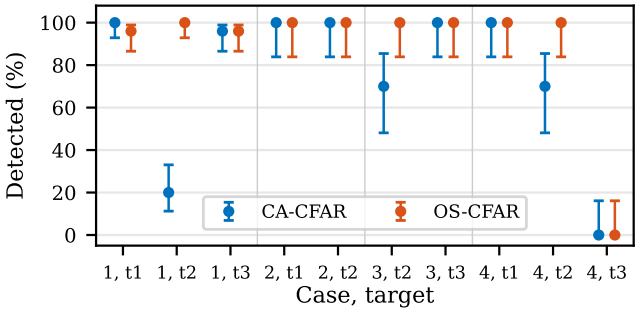


Fig. 4. Per-target detection rate of Table V (Appendix C) with Wilson 95 % intervals, CA-CFAR blue, OS-CFAR orange; 50 paired trials in case 1, 20 in cases 2–4. In case 4, "4, t1" is the strong target; the weak one is not separated.

VII. Limitations and Future Work

The unmodulated 100 μs pulse fixes the range resolution and blind range at 15 km; only one CPI is processed; end-to-end results contain thermal noise only; neither CFAR's false-alarm rate is validated by a noise-only Simulink run; same-cell targets are not separated; the OS threshold is conservative; and velocities alias beyond ± 7.5 m/s. Four measurements are still open: the MTI improvement factor with the clutter path connected, the noise-only P_{fa} of the Simulink 1-D path, the minimum SNR for a reliable 8-drop trigger, and the cause of the 170-sample delay reading (Appendix B, issue 1). Future work: a linear-FM chirp, staggered PRFs, a probability-of-detection sweep against the Swerling curves, a noise-only false-alarm test, tracking, and a hardware implementation.

References

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[2] H. M. Finn and R. S. Johnson, "Adaptive detection mode with threshold control as a function of spatially sampled clutter-level estimates," *RCA Review*, vol. 29, no. 3, pp. 414–464, Sep. 1968.

[3] H. Rohling, "Radar CFAR thresholding in clutter and multiple target situations," *IEEE Transactions on Aerospace and Electronic Systems*, vol. AES-19, no. 4, pp. 608–621, Jul. 1983.

[4] M. A. Richards, *Fundamentals of Radar Signal Processing*, 2nd ed. New York, NY, USA: McGraw-Hill Education, 2014.

[5] M. I. Skolnik, *Introduction to Radar Systems*, 3rd ed. New York, NY, USA: McGraw-Hill, 2001.

[6] F. J. Harris, "On the use of windows for harmonic analysis with the discrete Fourier transform," *Proceedings of the IEEE*, vol. 66, no. 1, pp. 51–83, Jan. 1978.

[7] E. B. Wilson, "Probable inference, the law of succession, and statistical inference," *Journal of the American Statistical Association*, vol. 22, no. 158, pp. 209–212, 1927.

Supplementary Material

Appendix A
Extended Derivations

Radar equation. Power P_t spread over a sphere and focused by gain G gives power density $P_tG/(4\pi R^2)$ at the target; a target of RCS σ reradiates $\sigma P_tG/(4\pi R^2)$, which spreads again over $4\pi R^2$ and is collected by an aperture $G\lambda^2/(4\pi)$. The product, divided by the losses L, is (3). At 20 km: $P_r = 1.79\times10^{-12}$ W (-87.5 dBm), $V_{rx} = 1.34\times10^{-6}$. Doubling the range costs $40\log_{10}2 = 12.0$ dB.

Matched-filter gain. With signal samples $s[k]$ and white noise of variance σ_n^2, the output SNR is $|\sum h[k]s[k]|^2/(\sigma_n^2\sum|h|^2)$. By the Cauchy–Schwarz inequality it is largest for $h \propto s^*$, giving N (20 dB for $N = 100$) for a constant-amplitude pulse. For the Hamming taps $(\sum w)^2/\sum w^2 = 53.54^2/39.35 = 72.9$ (18.6 dB); the peak power falls by 5.4 dB and the noise power by 4.0 dB.

CA-CFAR. The CUT power u is exponential with mean μ and the training sum $Z = \sum_{i=1}^{N} u_i$ is Gamma(N,μ). Conditioned on Z, $P(u > \alpha Z/N) = e^{-\alpha Z/(N\mu)}$, and averaging over Z with its moment-generating function gives $(1+\alpha/N)^{-N}$, independent of μ [2]. If only m training cells contain noise while the divisor stays N, the same steps give $(1+\alpha/N)^{-m}$.

OS-CFAR. With the k-th of N ordered exponential samples as the estimate, $P_{fa} = \prod_{j=N-k+1}^{N} j/(j+\alpha)$ [3]. Solving for 10^{-6} with $N = 626$, $k = 470$ gives $\alpha = 10.13$.

Pulse-pair. Successive pulses see the range shrink by v PRI, so the two-way phase advances by $\phi = 2\pi\cdot 2v\,\mathrm{PRI}/\lambda$. Multiplying by the conjugate of the previous pulse cancels the unknown initial phase, and $\arg(\cdot)\in(-\pi,\pi]$ gives $|v| \le \lambda/(4\,\mathrm{PRI})$.

MTI. $y[n] = x[n] - x[n-N_{\mathrm{PRI}}]$ applied to $Ae^{j2\pi f_d nT_s}$ gives $Ae^{j2\pi f_d nT_s}(1-e^{-j2\pi f_d \mathrm{PRI}})$, whose magnitude is $2|A\sin(\pi f_d \mathrm{PRI})|$.

Swerling I. If U is uniform, $-\ln U$ is exponential with unit mean, so $\sigma_k(-\ln U)$ is an exponential RCS with mean σ_k; the 10^{-4} lower limit caps it at $9.2\sigma_k$ (one draw in 10^4).

Wilson interval. For k successes in n trials, $\hat p = k/n$ and $z = 1.96$: $\left[\hat p + z^2/(2n) \pm z\sqrt{\hat p(1-\hat p)/n + z^2/(4n^2)}\right]/(1+z^2/n)$ [7].

Appendix B
Design Issues and Resolutions

Table IV lists every issue with its cause, fix and evidence. Items marked "not measured" are listed as open measurements in Sec. VII.

Appendix C
Supplementary Figures and Case Tables

Figs. 5 and 6 were moved here from Sections II and IV, Figs. 7–10 are the analytical figures referred to in the main text, Fig. 11 is the power map behind the detections of Fig. 2, and Figs. 12–14 are scope captures for the Echo, 1-D CFAR and Detection blocks. Table V gives the per-target results of the four CFAR test cases. Scenes: case 1, ranges 20/40/70 km, 2/3/4 m/s, RCS 1/1/1; case 2, 40/56 km, $+5/-3$ m/s; case

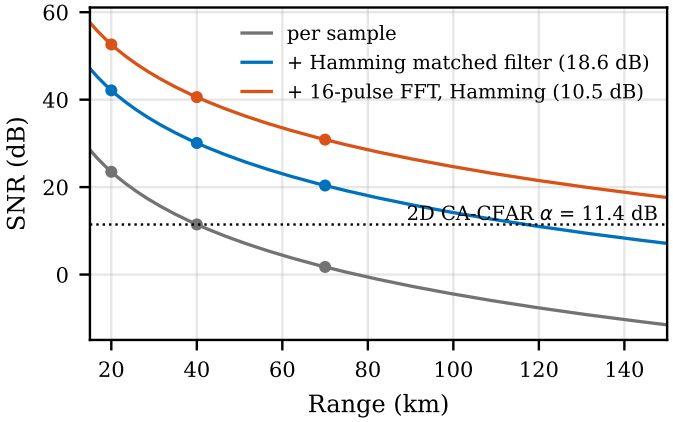


Fig. 5. SNR versus range for a 1 m^2 target (analytical, mean RCS). Dots mark the three simulated ranges; the dotted line is the 2-D CA-CFAR multiplier.

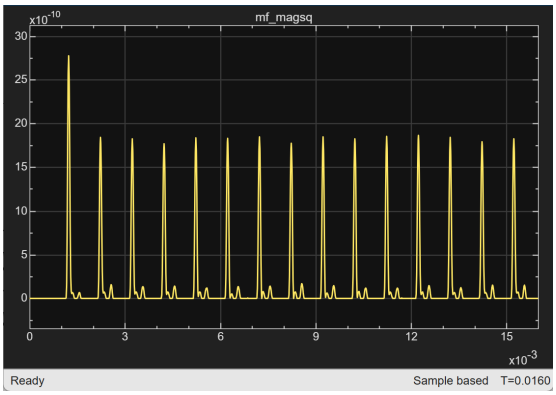


Fig. 6. Matched-filter power $|y|^2$ (`mf_magsq`) for three targets over one CPI. Target 1 dominates; targets 2 and 3 are the small peaks that follow. The first peak (PRI 2) is about 1.56 times the later ones because the MTI has no previous echo to subtract.

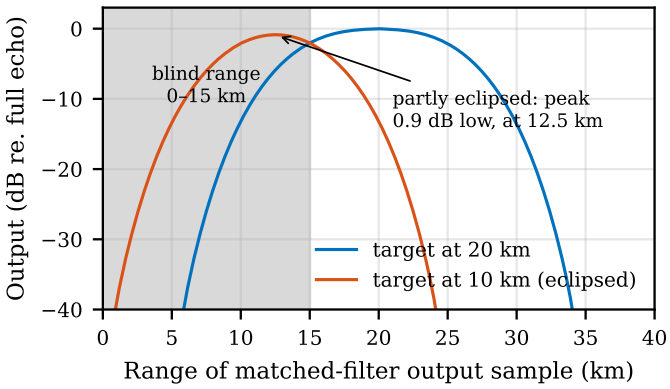


Fig. 7. Hamming matched-filter output for targets at 20 km and 10 km with the receiver blanked during transmission (analytical; blanking is not modelled in Simulink). The eclipsed echo appears 0.9 dB low and displaced.

TABLE IV
DESIGN ISSUES AND RESOLUTIONS

#	Problem	Cause	Fix	Evidence
1	Range extraction broke; a measured delay of 170 samples matched neither 100 (stale comment) nor 133	Matched filter changed from ones to the Hamming `range_window` for the 2-D map; 1-D CFAR and peak logic had been tuned for the triangle	Reverted for the 1-D path (19.95 km restored); the model now uses Hamming with a 199-sample guard and the 99-sample peak	07 Range README. A symmetric window cannot move the peak, so the cause of the 170 reading is not measured
2	200 m/s read as 5.03 m/s, suspected bug	Correct aliasing; within-pulse and between-pulse timescales conflated	None needed; $200 \bmod 15 = 5$	08 Velocity README
3	Mismatch sweep gave -24 dB at 200 m/s	`mf_magsq` is already $\lvert y\rvert^2$; the script squared it again	`max(abs(mf))`	Halved values within 0.46 dB of the rectangular curve; new Hamming sweep within 0.17 dB
4	Echo 2 dB stronger than budget	`radar_const` computed before L was set	Line moved after L	5 m/s peak 8.13×10^{-9} matched $L = 1$
5	Clutter inflated the CA-CFAR threshold and masked targets	Correlated clutter raises the local mean	MTI added; clutter block left unconnected	03 Noise README
6	Echo one PRI late; first PRI empty	Buffer must fill before the FFT	Whole-PRI delay drops out; offset stays 223 (1223 for a non-resetting counter)	Pure Echo scope: echoes at about 1.1, 1.3, 1.5 ms
7	Multiple triggers on one wide return	199-sample hump keeps the mask high	8-drop test, 150-sample lockout	Detection scope; weak-target threshold SNR not measured
8	Middle target missed by CA-CFAR	Training rows 251–266 sit 18–33 rows from the stronger target 1	OS-CFAR	Case 1: 10/50 versus 50/50
9	CA velocity bias +0.27 m/s	Mask cut on one side pulls the centroid (not investigated further)	OS-CFAR (bias 0.00 m/s)	Cases 3 and 4
10	Case 4 “20/20 and 0/20”	One cluster at row 367 matched to the first of two equidistant targets	Interpretation; same-cell separation is future work	Velocity error +1.85 m/s: the cluster is the 6.9 m/s target
11	Close targets undetectable; Doppler wrap ignored	Edge rows skipped; linear Doppler axis; $G_r = 200$	$G_r = 100$, G_d widened to 3, padding, circular Doppler	$G_r = 20$ split detections; $G_r = 250$ left 266 rows untested at each edge
12	Range and velocity blocks report one value	Built for a single return; hold the latest trigger	Path B (2-D map, CFAR, clustering)	Three targets reported in Table I
13	Monte Carlo $P_d = 100\,\%$	One SNR far above threshold; one pixel; no false alarms counted	Pd–SNR sweep with false-alarm count (future)	100/100 bounds P_d above 97.0 % (one-sided, Clopper–Pearson)
14	Gain block rejected a non-finite value	R^{-4} with $R \approx 0$: 1.2×10^{68} at $R = \epsilon$ exceeds 3.4×10^{38}	1 m floor; $\sigma = 0$ means off; `assert`	`init.m` lines 13 and 49

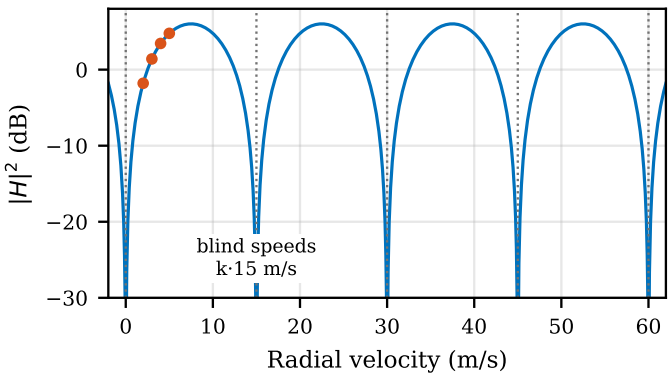


Fig. 8. Two-pulse MTI power response (analytical). Dots: targets at 2, 3, 4 and 5 m/s.

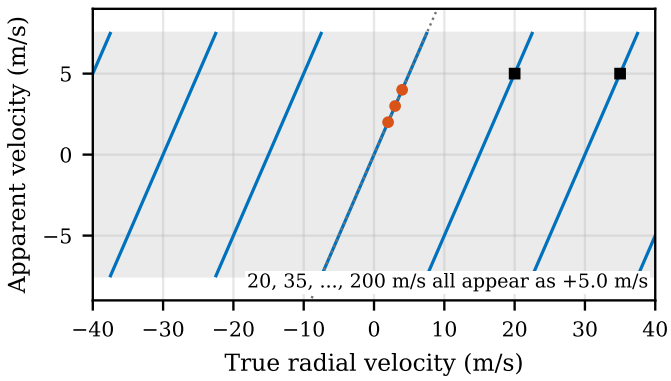


Fig. 9. True versus apparent velocity for PRI = 1 ms (analytical). The shaded band is the unambiguous interval ± 7.5 m/s.

3, 56/40 km, $-3/+6.9$ m/s, RCS 2/2; case 4, 40/56/40 km, $+5/-3/+6.9$ m/s, RCS 0.01/2/2.

Repeatability. Over 20 seeds the strongest-cluster headline was target 1 in 11 runs and target 2 in 9, because Swerling fading decides which is strongest; range was exact in every run. The script’s pass test (one cluster at 19.95 km) dates from the single-target model and fails by design.

APPENDIX D
PARAMETER TABLE

Table VI collects the parameters from `init.m` and the model blocks, with the derived values recomputed in `checks.py`; Table VII, moved here from Section II, lists the system limits derived from them.

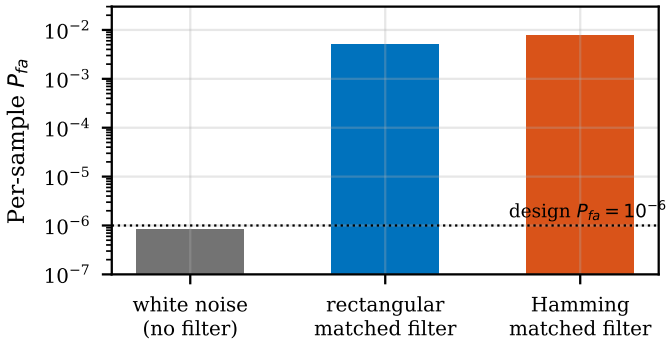


Fig. 10. Noise-only Python mirror of the 1-D CA-CFAR ($N = 32$, 100 guard cells per side, $\alpha = 17.28$; 4×10^7 white and 8×10^6 filtered samples). Correlated matched-filter noise raises P_{fa} by more than three orders of magnitude (analytical prediction, not a Simulink measurement).

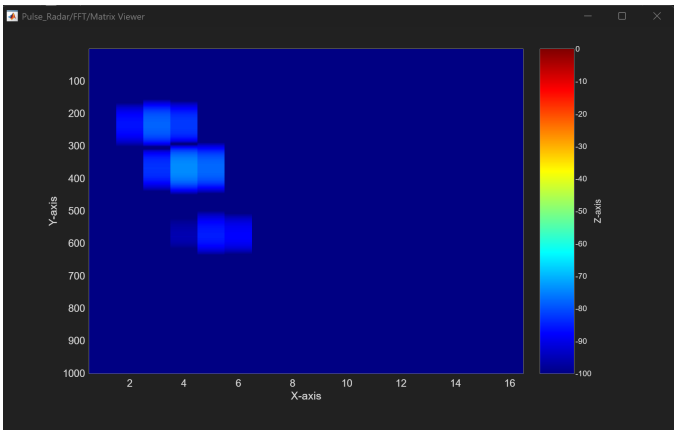


Fig. 11. Range–Doppler power map (dB) for the three-target baseline.

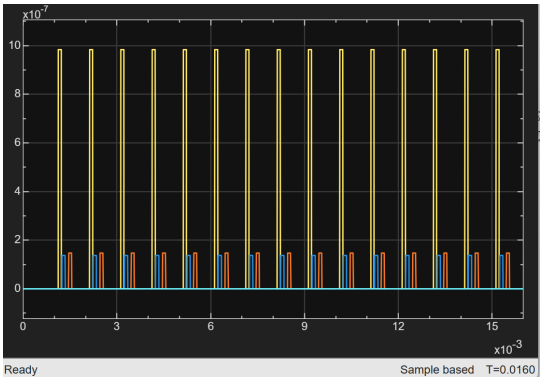


Fig. 12. Echo scope for targets at 20, 40 and 70 km with Swerling I fluctuation. The first PRI is empty and each echo starts one PRI after its nominal delay.



Fig. 13. 1-D CA-CFAR: square-law signal and adaptive threshold (earlier single-target capture, time axis in units of 10^{-4} s).

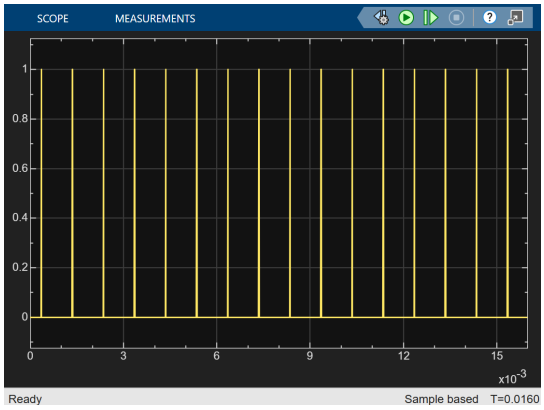


Fig. 14. Detection block output: one stretched trigger per PRI (earlier single-target capture).

TABLE V
PER-TARGET RESULTS, CASES 1–4 (FOUND; VELOCITY ERROR IN M/S)

Case, target	Row, v	CA	OS
1, t1	233, 2	50/50; +0.003	48/50; +0.014
1, t2	367, 3	10/50; +0.049	50/50; −0.010
1, t3	567, 4	48/50; +0.011	48/50; +0.016
2, t1	367, 5	20/20; +0.002	20/20; 0.000
2, t2	473, −3	20/20; −0.004	20/20; +0.001
3, t2	473, −3	14/20; +0.274	20/20; 0.000
3, t3	367, 6.9	20/20; −0.018	20/20; −0.004
4, t1	367, 5	20/20; +1.832	20/20; +1.852
4, t2	473, −3	14/20; +0.274	20/20; −0.001
4, t3	367, 6.9	0/20	0/20

TABLE VI
MODEL PARAMETERS AND DERIVED VALUES

Parameter	Value
f_c, λ	10 GHz, 0.03 m
P_t, G, L, F	100 kW, 35 dB, 2 dB, 3 dB
τ_p, PRI, T_s	100 μs, 1 ms, 1 μs (100, 1000 samples)
N_{CPI}, stop time	16, 16 ms
Ranges, N_{delay}	20/40/70 km; 133/267/467
Velocities	2/3/4 m/s (tests); `init.m`: 2/3/5
f_d	133.3/200/266.7 Hz (333.3 Hz at 5 m/s)
P_n, $\sigma^2_{I,Q}$	7.99×10^{-15} W, 3.99×10^{-15}
Clutter variance (unconnected)	$50\times$ thermal, 17.0 dB
1-D CFAR	$N = 32$, guard 200, $\alpha = 17.28$, 233 taps, delay 116
Detection	8 drops, lockout 150, stretch 10
Latency	$99 + 116 + 8 = 223$; echo chain +1000
2-D CA-CFAR	$T_r, T_d, G_r, G_d = 16, 1, 100, 3$; $N = 690$; $\alpha = 13.955$
2-D OS-CFAR	$G_d = 2$; $N = 626$; $k = 470$; $\alpha = 14.02$
Doppler window	Hamming(16): -39 dB sidelobe, 1.54 dB loss
Pulse-pair gain	2.387 m/s per rad

TABLE VII
DERIVED SYSTEM LIMITS (COMPUTED)

Quantity	Formula	Value
Pulse repetition frequency	$1/\mathrm{PRI}$	1 kHz
Duty cycle, mean power	τ_p/PRI, $P_t\tau_p/\mathrm{PRI}$	10 %, 10 kW
Unambiguous range	$c\,\mathrm{PRI}/2$	150 km
Blind range	$c\tau_p/2$	15 km
Range resolution	$c\tau_p/2$	15 km
Range-bin spacing	$cT_s/2$	150 m
Noise bandwidth	$1/T_s$	1 MHz ($\approx$100$\times$ pulse)
Matched-filter gain	$(\sum h)^2/\sum h^2$	20.0 dB rect., 18.6 dB Hamming
Doppler shift	$2v/\lambda$	66.7 Hz per m/s
Unambiguous velocity	$\pm\lambda/(4\,\mathrm{PRI})$	$\pm$7.5 m/s
Velocity resolution	$\lambda/(2N\,\mathrm{PRI})$	0.9375 m/s
MTI blind speeds	$k\lambda/(2\,\mathrm{PRI})$	$k\times$15 m/s